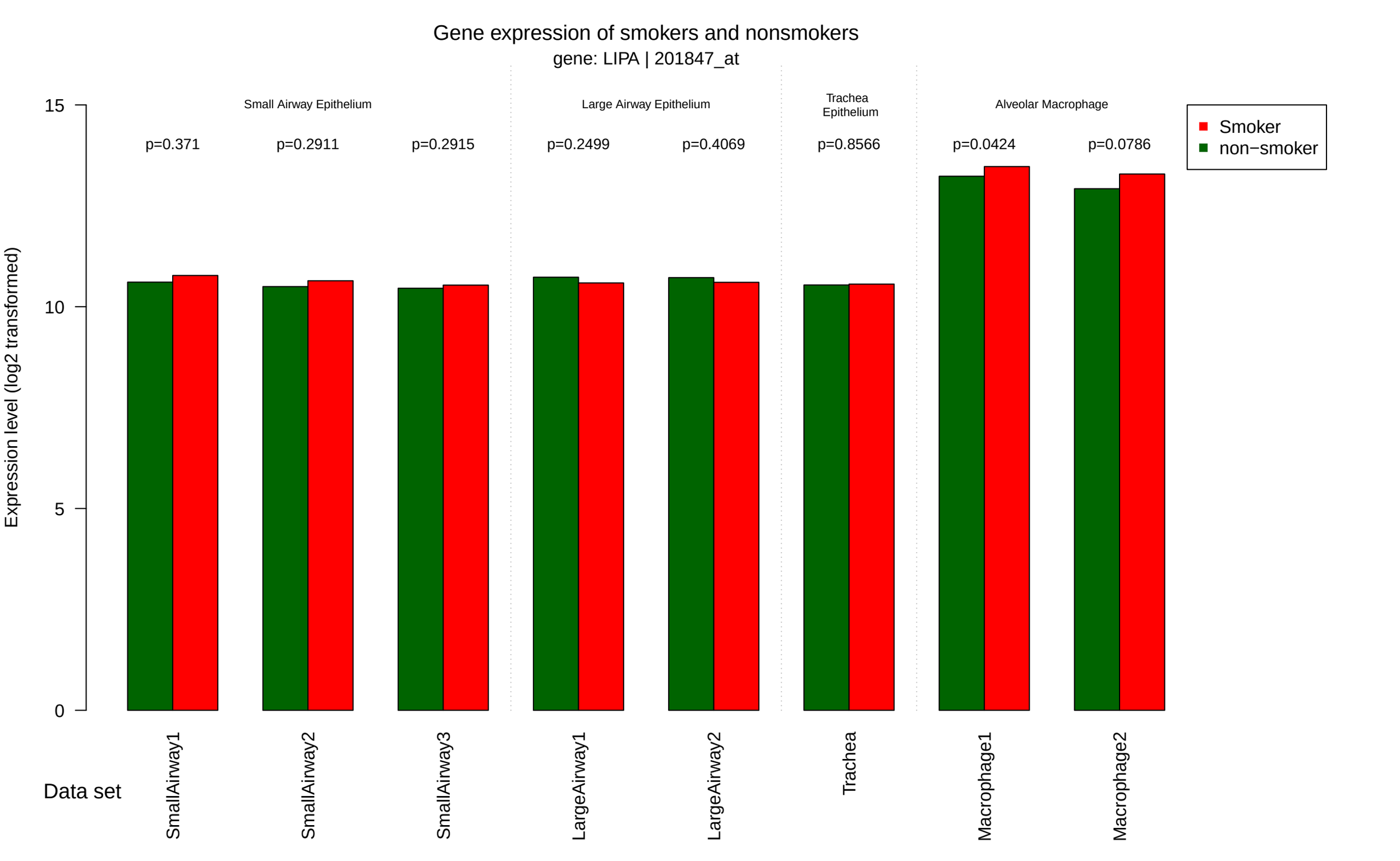
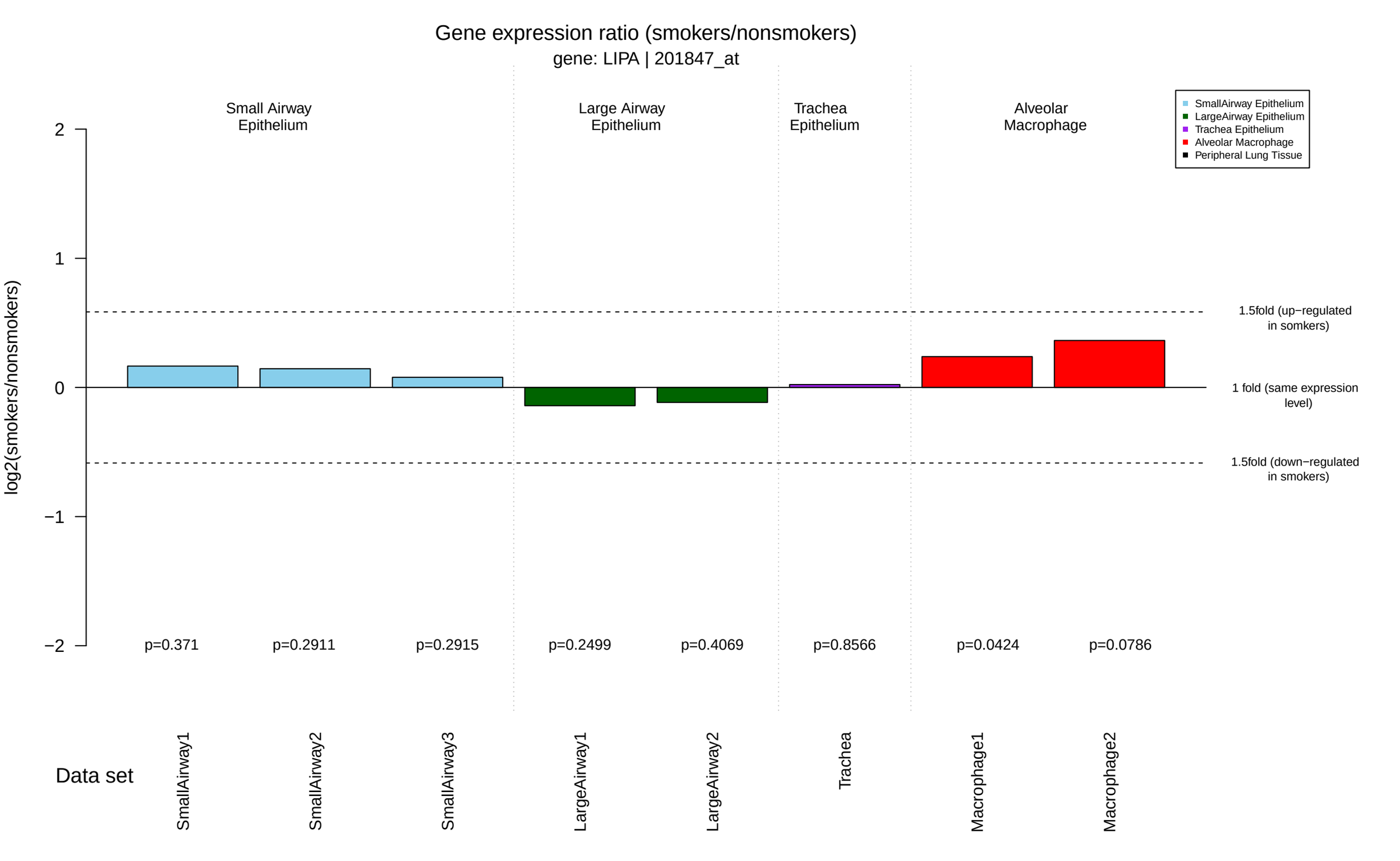
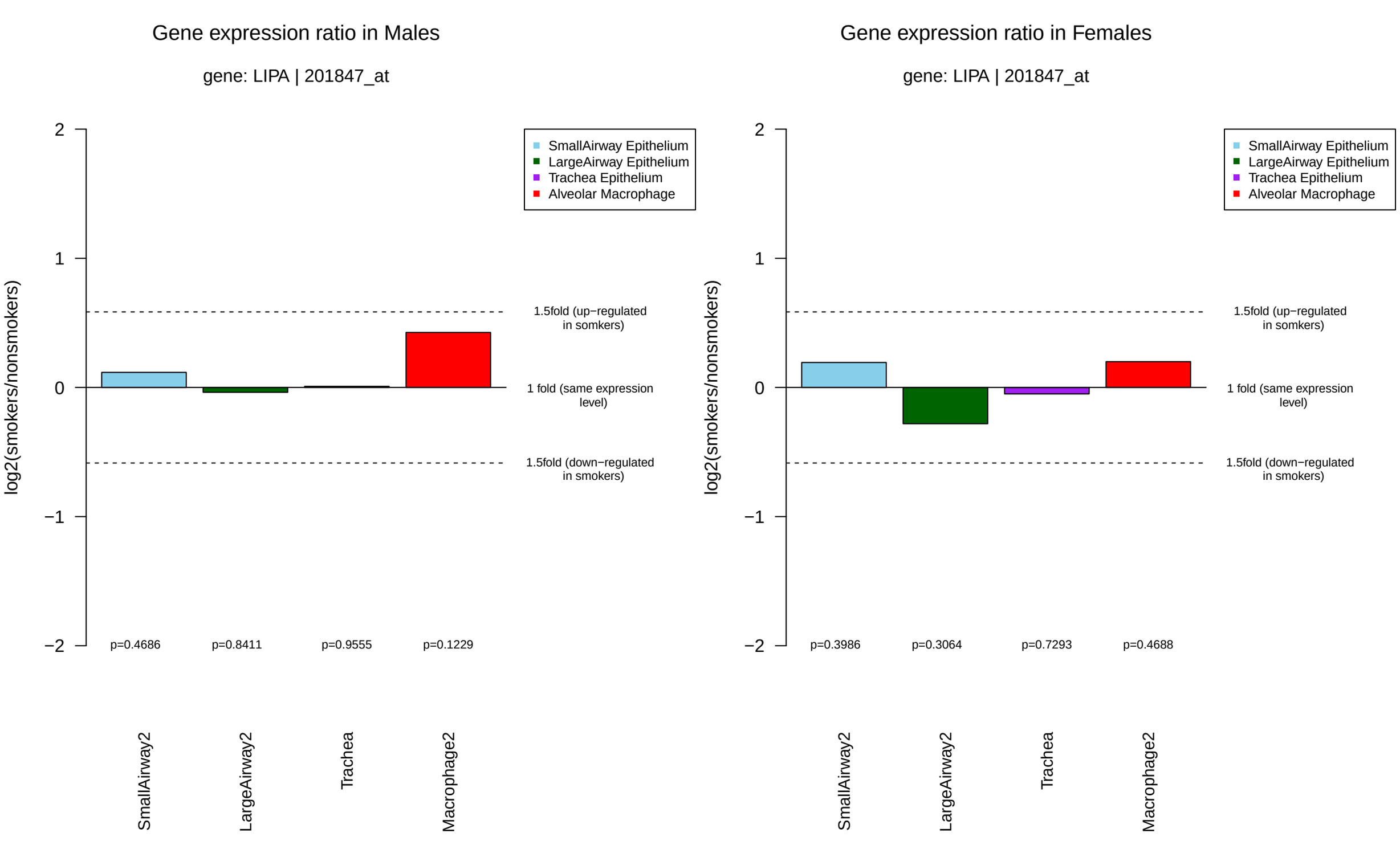
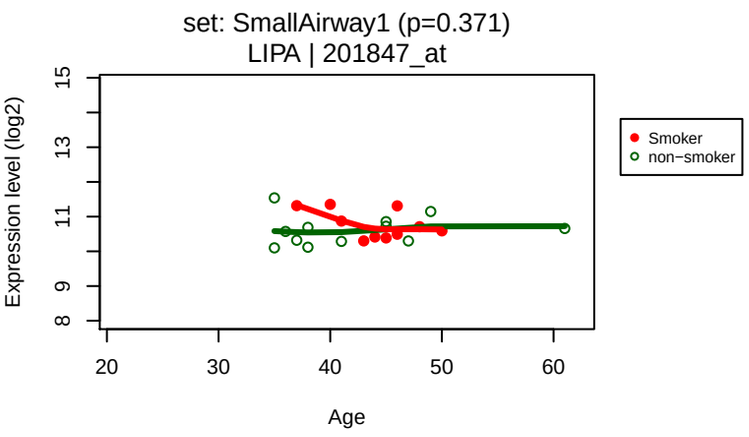
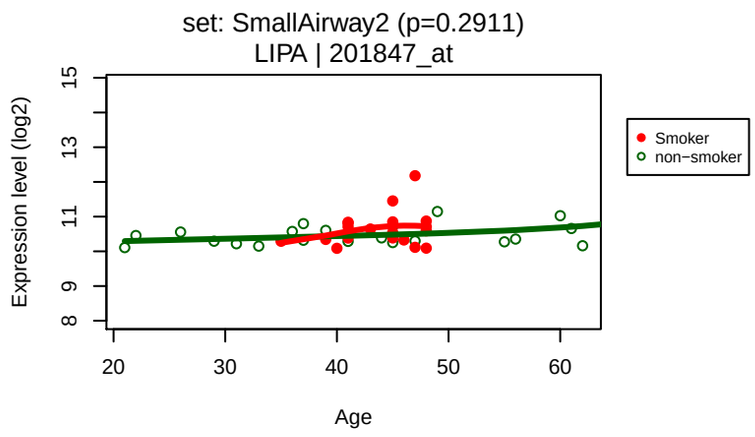
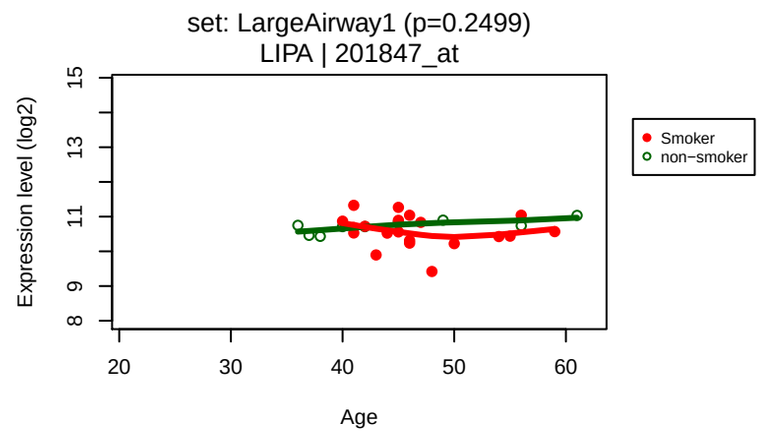

Gene expression of smokers and nonsmokers
gene: LIPA | 201847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.371
p=0.2911
p=0.2915
p=0.2499
p=0.4069
p=0.8566
p=0.0424
p=0.0786
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

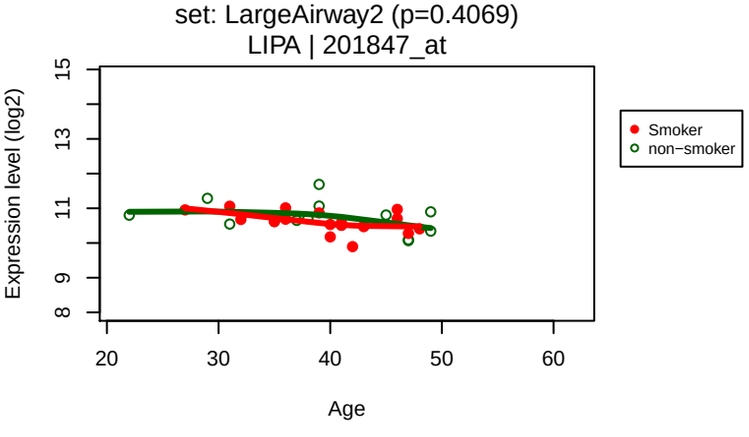

Gene expression ratio (smokers/nonsmokers)
gene: LIPA | 201847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.371
p=0.2911
p=0.2915
p=0.2499
p=0.4069
p=0.8566
p=0.0424
p=0.0786
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

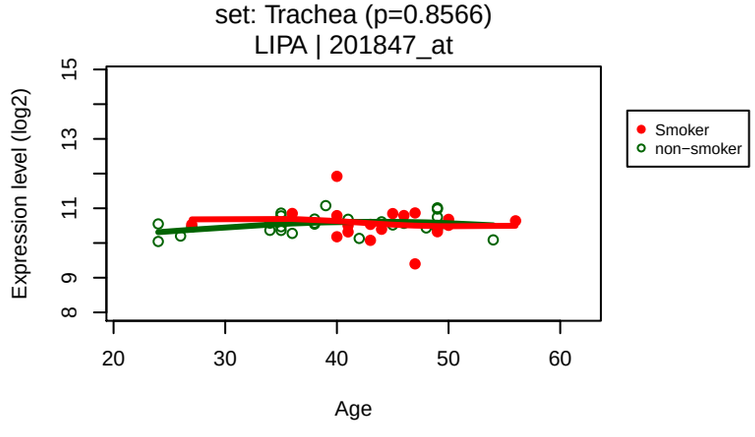

Gene expression ratio in Males
gene: LIPA | 201847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4686
p=0.8411
p=0.9555
p=0.1229
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LIPA | 201847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3986
p=0.3064
p=0.7293
p=0.4688
SmallAirway2
LargeAirway2
Trachea
Macrophage2

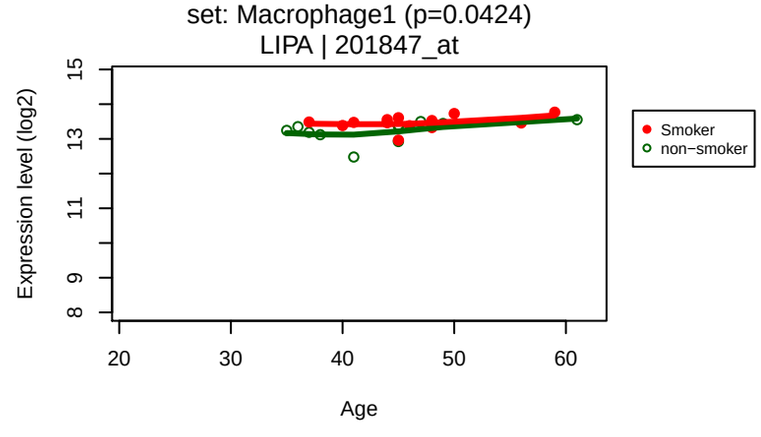

set: SmallAirway1 (p=0.371)
LIPA | 201847_at
Expression level (log2)
Age
Smoker
non−smoker

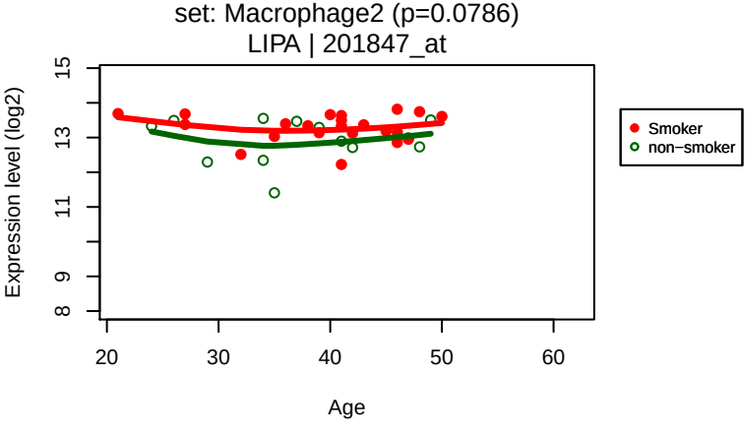

set: SmallAirway2 (p=0.2911)
LIPA | 201847_at
Expression level (log2)
Age
Smoker
non−smoker

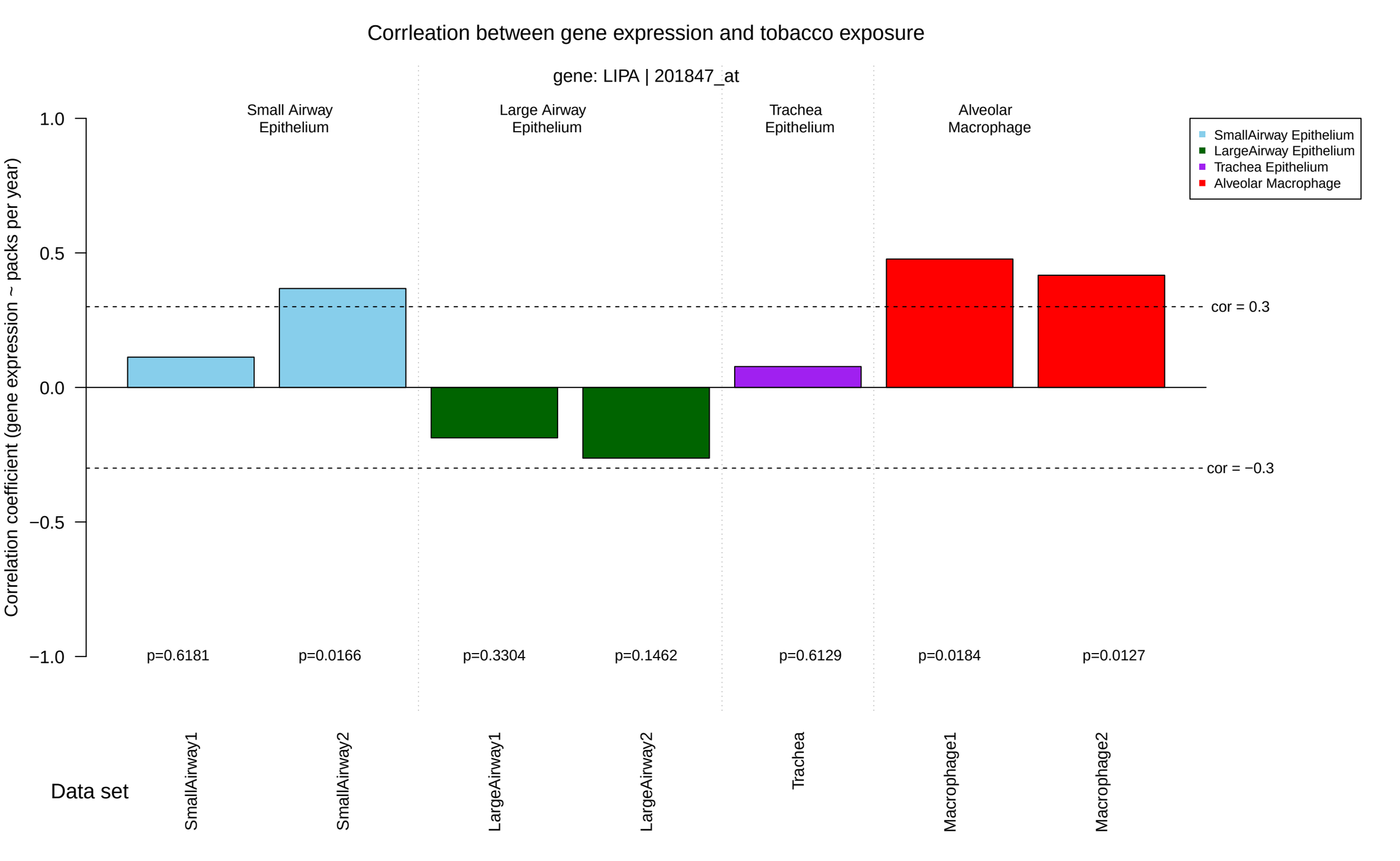

set: LargeAirway1 (p=0.2499)
LIPA | 201847_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4069)
LIPA | 201847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8566)
LIPA | 201847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0424)
LIPA | 201847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0786)
LIPA | 201847_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LIPA | 201847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6181
p=0.0166
p=0.3304
p=0.1462
p=0.6129
p=0.0184
p=0.0127
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2